

MEZBAUR RAHMAN

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RESEARCH INTERESTS

Large Language Models | Vision Language Models | Learning from Noisy Labels | Semi Supervised Learning

EDUCATION

University of Illinois Chicago

Ph.D. in Computer Science and Engineering
Current CGPA: **4.00** out of 4.00

Chicago, United States

Aug 2023 - Present

Islamic University of Technology

M.Sc in Computer Science and Engineering
CGPA: **3.96** out of 4.00

Gazipur, Bangladesh

Jan 2020 - June 2023

Islamic University of Technology

B.Sc in Computer Science and Engineering
CGPA: **3.86** out of 4.00 (4th in class)

Gazipur, Bangladesh

Jan 2016 - Nov 2019

PROFESSIONAL EXPERIENCE

University of Illinois Chicago

Graduate Research Assistant — advised by [Prof. Cornelia Caragea](#)
Department of Computer Science, ~~dl~~nlp space Lab

Chicago, United States

Aug 2023 – Present

- Conducting research at the intersection of Natural Language Processing and Machine Learning, focusing on semi-supervised learning and learning from noisy labels using guidance from Large Language Models (LLMs) and Vision-Language Models (VLMs).
- Developed novel frameworks that leverage LLMs for label refinement, pseudo-labeling, and historical training dynamics-aware co-training, improving overall text classification performance.

Islamic University of Technology

Lecturer, Department of Computer Science and Engineering

Gazipur, Bangladesh

Jan 2020 – Jul 2023

- Taught and designed undergraduate courses on Data Structures, Algorithms, and Database Systems; supervised research and capstone projects in Data Mining and NLP.

Samsung R&D Institute Bangladesh

Software Engineering Intern (Mobile Application Group)

Dhaka, Bangladesh

Nov 2018 – Jan 2019

- Worked on developing an iOS app that simulated dynamic light projection and shading effects on 3D objects using OpenGL ES and native iOS rendering classes.

PUBLICATIONS

- **Md Mezbaur Rahman** and Cornelia Caragea. 2025. *LLM-Guided Co-Training for Text Classification*. In **Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025)**, pages 31092–31109, Suzhou, China. Association for Computational Linguistics.
 - Proposed an LLM-guided weighted co-training framework for semi-supervised text classification.
 - Achieved state-of-the-art results on 4/5 benchmarks and ranked first among 14 SSL methods.
- Mohsinul Kabir, Tasnim Ahmed, **Md Mezbaur Rahman**, Polydoros Giannouris, and Sophia Ananiadou. 2025. *Semantic Label Drift in Cross-Cultural Translation*. arXiv preprint. Submitted to the **Language Resources and Evaluation Conference (LREC 2026)**.
 - Investigated semantic label drift during translation across culturally diverse languages.

- Proposed alignment-aware evaluation to mitigate cross-cultural label inconsistency.
- **Md Mezbaur Rahman**, Mohammed Saidul Islam, Md Tahmid Rahman Laskar, Md Azam Hossain, and Abu Raihan Mostofa Kamal. 2023. *Multihop Factual Claim Verification Using Natural Language Prompts*. In **Proceedings of the Canadian Conference on Artificial Intelligence (Canadian AI 2023)**. Canadian Artificial Intelligence Association (CAIAC), June 2023.
 - Developed prompt-based fine-tuning strategies for verifying multi-sentence factual claims.
 - Investigated how multi-hop reasoning increases complexity in claim verification.
- Syed Mohammed Sartaj Ekram, Adham Arik Rahman, Md Sajid Altaf, Mohammed Saidul Islam, Mehrab Mustafy Rahman, **Md Mezbaur Rahman**, Md Azam Hossain, and Abu Raihan Mostofa Kamal. 2022. *BanglaRQA: A Benchmark Dataset for Under-Resourced Bangla Language Reading Comprehension-Based Question Answering with Diverse Question-Answer Types*. In **Findings of the Association for Computational Linguistics: EMNLP 2022**, pages 2518–2532, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.
 - Introduced a 14K+ QA dataset from Bangla Wikipedia for reading comprehension.
 - Improved transformer-based QA performance for low-resource Bangla language.
- **Md Mezbaur Rahman**, Saadman Malik, Mohammed Saidul Islam, Fardin Saad, Md Azam Hossain, and Abu Raihan Mostofa Kamal. 2022. *An Efficient Approach to Automatic Tag Prediction from Movie Plot Synopses Using Transformer-Based Language Model*. In **Proceedings of the 25th International Conference on Computer and Information Technology (ICCIT 2022)**, pages 501–505. IEEE.
 - Designed a transformer-based model for predicting movie tags from plot synopses.
 - Compared transformer models with LSTM and baseline neural architectures.
- Mohammed Saidul Islam, Iqram Hussain, **Md Mezbaur Rahman**, Se Jin Park, and Md Azam Hossain. 2022. *Explainable Artificial Intelligence Model for Stroke Prediction Using EEG Signal*. **Sensors**, 22(24):9859. MDPI.
 - Applied explainable machine learning models for EEG-based stroke prediction.
 - Improved interpretability using model-level and feature-level XAI methods.

SKILLS

ML Frameworks	PyTorch, Transformers, RLHF, DeepSpeed, vLLM, Hugging Face, LoRA
Programming Languages	Python, C/C++, Java, Bash/Shell
Data Science Libraries	NumPy, Pandas, Matplotlib, Scikit-learn
Tools & Systems	Git, Docker, Kubernetes, LaTeX/Overleaf, MySQL, Oracle SQL

ONLINE CERTIFICATION

- **Neural Networks and Deep Learning** by DeepLearning.AI on Coursera
- **Natural Language Processing with Probabilistic Models** by DeepLearning.AI on Coursera
- **Natural Language Processing with Classification and Vector Spaces** by DeepLearning.AI on Coursera

SIGNIFICANT PROJECTS

- **Target Invariant Stance Detection Using Large Language Models** ([GitHub Link](#))
 - Explored the performance of Large Language Models (LLMs) in stance detection under zero-shot and fine-tuned settings.
 - Compared 7B-parameter models against smaller fine-tuned models to analyze trade-offs between model size, contextual reasoning, and generalization.
- **LLM-Guided Node Classification in Semi-Supervised Settings** ([GitHub Link](#))
 - Integrated LLM-based sentence encoders with Text-Attributed Graphs to enrich node representations.
 - Employed LLM-guided pseudo labels to initialize semi-supervised learning, improving performance with increasing unlabeled proportions.